Fisher Geometry of a Live Decision-Model API: Ticket Triage Sensitivity under TypeSafe

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Abstract

Decision-model APIs expose calibrated categorical distributions over operational questions—request type, escalation need, impact—conditioned on structured ticket state. Practitioners routinely treat those distributions as point predictions; less often do they ask how *sensitive* the reported law is to the continuous parameters that define the ticket (dollar amount, wait time). This paper studies that sensitivity for a live TypeSafe decision model using the Jevometry geometry stack: finite-difference Jacobians, Fisher information on each node, entropy, and an explicit capability matrix that refuses unsupported claims.

We first show that an aggressive finite-difference stencil (h_{amount}=1.0, h_{wait}=0.25 with a two-scale stability check) marks all three live node geometries `unstable/step_stability_exceeded`, so Fisher traces cannot be trusted. Shrinking steps to 0.25 and 0.05 and using a single-scale live stencil restores `ok` Jacobian and Fisher status on every node while preserving the qualitative finding that wait-time diagonals dominate amount. A fresh live rerun (75 traces, zero failures) yields centre-point entropies $H \approx 1.04/0.50/0.31$ nats for `request_type/needs_escalation/impact`, with Fisher traces $\mathrm{tr}(I) \approx 0.86/0.085/2.61$. Two nodes remain rank-deficient (`not_identifiable`), and system/action Fisher stay `unsupported` because no joint model is declared and routing is deterministic. An offline analytic twin on a 3×3 grid provides a noise-free contrast. Together, the study is a template for honest geometry of commercial decision APIs: report entropy when Fisher is unstable, shrink stencils before claiming sensitivity, and separate node geometry from joint or policy claims. Code, CSV summaries, and figures ship with Jevometry (`papers/typesafe-ticket/`).

1 Introduction

Ticket triage is a canonical multi-question decision workflow. Given a structured ticket—amount in USD, hours already waiting, channel, free-text summary—a decision model returns categorical distributions over (i) request type (billing / technical / refund), (ii) whether escalation is needed, and (iii) impact severity. Production systems then map those answers into a deterministic routing label (standard queue, priority, refund review, escalate). Large language models and specialized decision APIs have made such distributions cheap to obtain [Yao et al., 2023, Park et al., 2023, TypeSafe, 2026], but cheap does not mean geometrically well-behaved: finite-difference estimates of $\partial_\theta P$ can fail stability checks under API noise, and without a declared joint the three marginals do not determine a system Fisher matrix [Jevometry Contributors, 2026].

Information-theoretic multi-agent sensing and Dec-POMDP active perception optimize mutual information, prediction rewards, or belief-dependent payoffs [Lauri and Oliehoek, 2020, Lauri et al., 2020, Oliehoek and Amato, 2016, Shi et al., 2026, Molloy and Nair, 2023, Corah and Michael, 2021]. Those lines choose *what to observe*. A companion paper studies *which coordination rule* retains Fisher information after action aggregation [Li, 2026]. Here the complementary question is narrower and more operational: *given a fixed live decision model and*

a fixed question set, how sensitive are the reported node distributions to ticket parameters, and when is that sensitivity trustworthy?

Contributions.

1. We document a live TypeSafe ticket experiment in Jevometry, including the capability matrix that correctly marks system/action Fisher and likelihood-based inference as unsupported.
2. We show that stencil choice is first-order: the original live steps yield `step_stability_exceeded` on all nodes; shrinking amount and wait steps and switching live to a single-scale stencil restores `ok` node geometry.
3. We report reproducible centre-point entropy and Fisher summaries from a fresh live re-run, with wait-hours diagonals dominating amount, plus an offline synthetic grid for noise-free comparison.
4. We state limitations honestly: rank deficiency on binary/score nodes, no joint model, deterministic policy, and API non-determinism across re-runs.

2 Related work

Active perception and Dec-POMDPs. Prediction-reward formulations [Lauri and Oliehoek, 2020] and policy-graph improvements for information gathering [Lauri et al., 2020] optimize belief-dependent objectives under decentralization. Classical POMDP planning emphasizes value of information under partial observability [Kaelbling et al., 1998, Oliehoek and Amato, 2016]. IMAS^2 adds joint agent selection under mutual-information objectives [Shi et al., 2026]. Our setting does not choose sensors; it *audits* a deployed question set.

Fisher geometry and DPI. Fisher information is the local Riemannian metric of a parametric family [Fisher, 1925, Amari, 2016, 1998]. Data-processing inequalities constrain pushforwards [Zamir, 1998, Cover and Thomas, 2006]. Sensor placement and mutual-information maximization provide algorithmic templates [Krause et al., 2008] but assume controllable observation design.

LLM / agent decision stacks. Tool-using and multi-agent LLM systems [Yao et al., 2023, Park et al., 2023] expose decision surfaces that look like categorical models. Geometry on those surfaces is rarely reported. TypeSafe is treated here as a black-box decision-model API accessed through an official Python SDK [TypeSafe, 2026]; we do not reverse-engineer model internals.

Companion coordination paper. Li [2026] hold analytic agent laws fixed and vary deterministic aggregation maps, ranking rules by Fisher loss after the action layer. The present paper holds the live API fixed and studies parameter sensitivity of the reported node laws—orthogonal axes of the same Jevometry programme [Jevometry Contributors, 2026].

3 Method

3.1 Ticket decision model

Let $\theta = (\text{amount}, \text{wait_hours}) \in \mathbb{R}^2$ with design centre $\theta_\star = (250, 8)$ (USD, hours). The live adapter renders a structured state and queries three TypeSafe questions:

- `request_type`: Choice on $\{\text{billing}, \text{technical}, \text{refund}\}$;

- `needs_escalation`: Noul (binary) on $\{\text{false}, \text{true}\}$;
- `impact`: Score on $\{0, 1, 2\}$ (low/medium/high).

Routing semantics are declared `deterministic_policy`: the production map from reported probabilities to an action label is not a sampling model, so trajectory/action Fisher are refused by the capability matrix.

3.2 Node Fisher via finite differences

For a node with reported probabilities $P_\theta \in \Delta^{K-1}$, Jevometry estimates a Jacobian $J \approx \partial_\theta P$ by centered finite differences along each parameter, optionally at multiple step scales, then forms

$$I(\theta) = J^\top \mathrm{diag}(P_\theta)^{-1} J \tag{1}$$

when regularity checks pass [Jevometry Contributors, 2026]. Entropy $H(P) = -\sum_k P_k \log P_k$ is always available from the reported distribution and does not require a stable Jacobian.

Definition 1 (Relative step stability)**.** *A multi-scale stencil compares Jacobians at step h and $h/2$. If the relative discrepancy exceeds a threshold (default 0.10), the geometry is marked* ***`unstable`*** *with reason* ***`step_stability_exceeded`****. Unstable Fisher values may still be printed as diagnostics but are not validated geometry.*

3.3 Stencil shrink for live APIs

Live API responses are noisy across repeats. Empirically, the original live defaults h_{amount}=1.0, h_{wait}=0.25 with `step_scales`= $[1.0, 0.5]$ failed the multi-scale check on every node. We adopt:

- h_{amount}=0.25, h_{wait}=0.05;
- live stencil `step_scales`= $[1.0]$ (single scale);
- offline synthetic keeps the two-scale stability stencil.

Repeats remain 5 with a live budget cap of 100 attempts; the acquisition plan expects 25 requests at the single centre point used for live.

3.4 Offline synthetic twin

The analytic `TicketSystemAdapter` defines softmax logits that depend smoothly on scaled amount and wait [Jevometry Contributors, 2026]. Offline runs a 3×3 grid amount $\in \{100, 250, 600\}$, wait_hours $\in \{2, 8, 30\}$ and reports node geometry plus a declared product *surrogate* action distribution (explicitly not the production routing law).

3.5 What we refuse to claim

Without a declared joint $P_\theta(Z_1, Z_2, Z_3)$, system Fisher is `unsupported/no_joint_model`. With deterministic routing, action Fisher is `unsupported/deterministic_policy`. Without a sampling contract, CRLB-style inference is refused. These refusals are features of the analysis object model, not bugs.

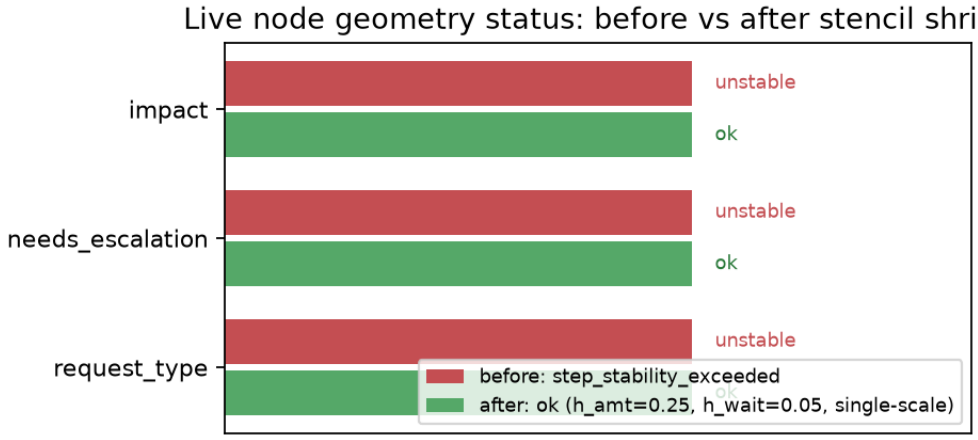


Figure 1: Live node geometry status before and after the stencil shrink (PR `fix/ticket-live-stencil-stability`). Before: all three nodes `unstable`/`step_stability_exceeded`. After: all `ok`.

Table 1: Live TypeSafe centre point $\theta_\star = (250, 8)$ after stencil shrink. Fisher status is `ok` on all nodes; identifiability remains rank-deficient for escalation and impact.

Node	H (nats)	$\mathrm{tr}(I)$	I_{amt}	I_{wait}	Ident.
request_type	1.040	0.858	0.138	0.720	—
needs_escalation	0.500	0.085	0.0225	0.0625	rank-def.
impact	0.309	2.613	0.154	2.460	rank-def.

4 Experiments

4.1 Protocol

All commands from the repository root:

```
uv sync --locked
uv run python examples/jev_ticket_sensitivity/run.py
# optional:
uv run python examples/jev_ticket_sensitivity/run.py --live
uv run python examples/jev_ticket_sensitivity/paper_export.py
```

Credentials: environment variables `TYPESAFE_API_KEY` and `TYPESAFE_MODEL` (values never committed). Live outputs under `examples/jev_ticket_sensitivity/output/` are gitignored; paper tables are summarized into `papers/typesafe-ticket/data/`.

4.2 Before vs after stencil shrink

Figure 1 summarizes the status flip. We deliberately do *not* treat pre-fix Fisher numbers as validated results.

4.3 Live centre-point results (fresh re-run)

A live re-run after the stencil fix completed with 75 traces, 0 failures, `incomplete=False`, run id `ticket-sensitivity-live-20261010T143812-0` (UTC timestamp in `summary.json`).

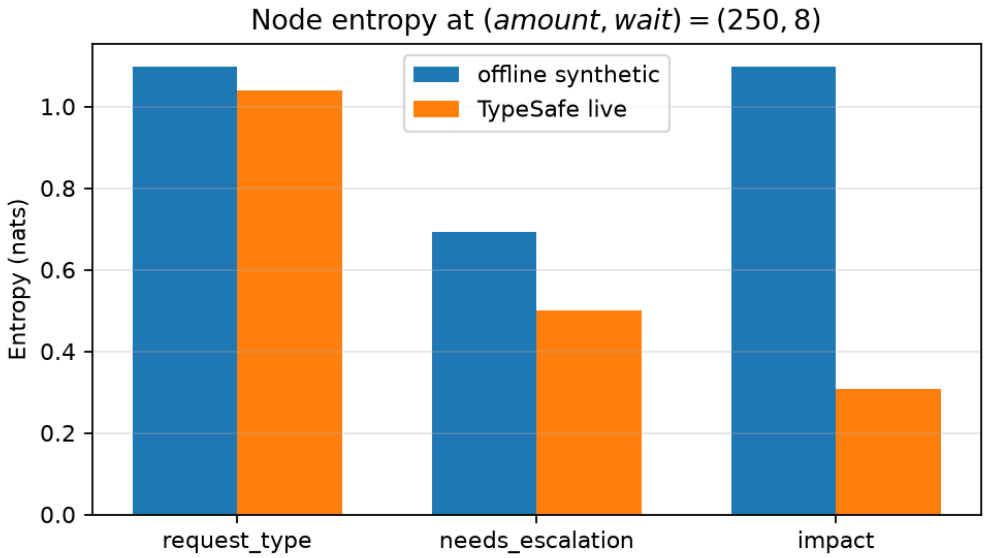


Figure 2: Entropy at the design centre: offline synthetic vs TypeSafe live.

Table 2: Capability matrix (identical structure for live and offline).

Capability	Status	Reason
distribution_comparison	ok	—
node_geometry	ok	—
system_information	unsupported	no_joint_model
action_information	unsupported	deterministic_policy
inference	unsupported	no_likelihood_model

Table 1 and Figures 2–3 show: (i) entropy is highest for `request_type` and lowest for `impact`; (ii) $\mathrm{tr}(I)$ is dominated by the wait-hours diagonal; (iii) binary/score nodes can be Fisher-ok yet `not_identifiable` (rank-deficient I), so local invertibility fails even when stability passes.

4.4 Capability matrix

4.5 Offline synthetic grid

At the shared centre, synthetic entropies are $H \approx 1.097/0.692/1.098$ with tiny Fisher traces ($\mathrm{tr}(I) \sim 10^{-4}$–$10^{-5}$). The live model is a different law: absolute Fisher scales are not comparable across providers, but the *status* and *relative* wait>amount pattern are.

5 Discussion

Stencil first, narrative second. Publishing Fisher numbers from an unstable stencil would overstate sensitivity. The capability/status channel is the right gate: entropy remains reportable; Fisher waits for `ok`.

Wait-time sensitivity. Across live nodes, $I_{\mathrm{wait}} \gg I_{\mathrm{amt}}$ at $\theta_\star$. Operationally, wait-time interventions move the reported distributions more than small amount perturbations at this centre—subject to the scale normalization in the ParameterSpec (100 USD, 12 hours).

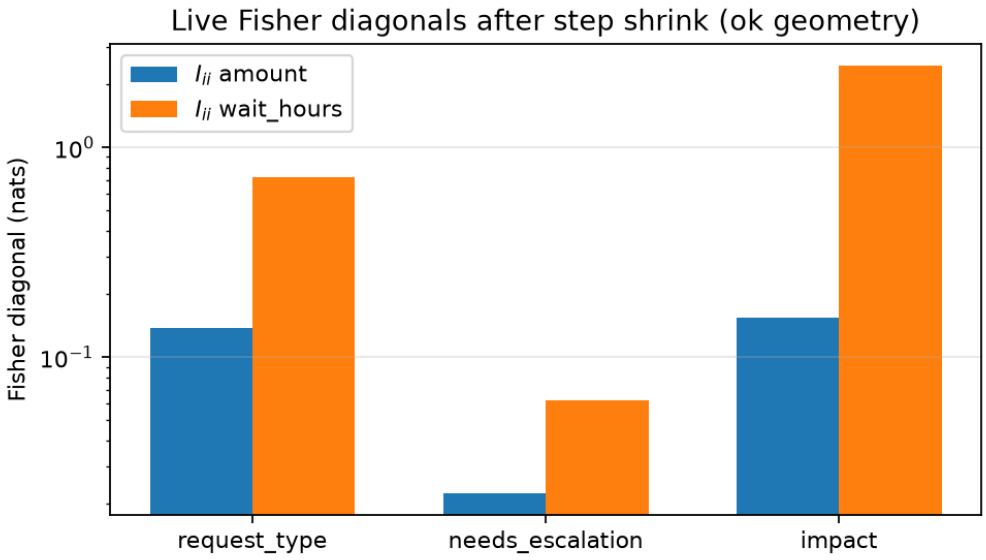


Figure 3: Live Fisher diagonals (log scale). Wait-hours dominates amount on every node after the stencil shrink.

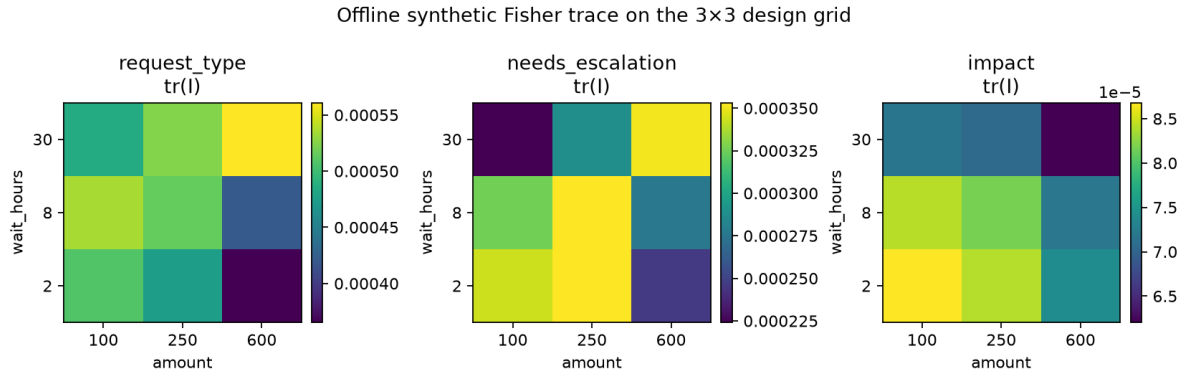


Figure 4: Offline synthetic tr(I) on the 3×3 design grid. Traces are small relative to live (analytic logits change slowly in raw parameter units) but remain `ok` under the two-scale stencil.

Identifiability vs stability. Stability says the finite-difference Jacobian is repeatable at the chosen h. Identifiability asks whether I has full parameter rank. Binary nodes with two-dimensional θ are structurally prone to rank deficiency; treating `ok` Fisher as "fully informative about both parameters" would be incorrect.

Relation to coordination loss. If a council later compresses these three nodes through a deterministic aggregator, Fisher loss after the action layer can be measured as in Li [2026]—but only after a joint is declared. The live ticket example currently stops at node geometry by design.

6 Limitations

- **Single live centre.** Budget limits live to $\theta_\star$; grid conclusions use the synthetic twin.
- **API non-determinism.** Re-runs change numerical Fisher values; we commit summaries, not raw `traces.jsonl`.

- **No joint / no action Fisher.** Deterministic routing and absent joint block system-level claims.
- **Black-box vendor model.** We do not claim internal TypeSafe training details; the object of study is the reported distribution interface.
- **Synthetic scale mismatch.** Analytic logits are slow in raw $(amount, wait)$ units, so absolute $\mathrm{tr}(I)$ is not a live baseline—only a methodological twin.

7 Conclusion

Live decision-model APIs can be placed under Fisher geometry if—and only if—finite-difference stencils are stable. For TypeSafe ticket triage in Jevometry, shrinking amount/wait steps and using a single-scale live stencil turns `step_stability_exceeded` into `ok` node geometry, reveals wait-dominated diagonals, and still surfaces rank-deficiency and capability refusals that keep joint and policy claims honest. Entropy remains the fallback when geometry fails. Future work: declare a product or tree joint for live nodes, attach a stochastic routing surrogate for action Fisher, and extend the live grid under a larger budget.

Reproducibility

Branch `paper/typesafe-ticket-sensitivity`; paper artifacts under `papers/typesafe-ticket/`; experiment code under `examples/jev_ticket_sensitivity/`. Offline needs no credentials; live requires `TYPESAFE_API_KEY` and `TYPESAFE_MODEL`.

A Experiment design details

Live acquisition. At $\theta_\star = (250, 8)$ with three nodes, five repeats, and a single-scale centered difference along two parameters, the acquisition planner expects 25 TypeSafe requests (budget cap 100). The re-run reported in Section 4 completed with 75 stored traces (including bookkeeping rows) and zero failures.

Parameter scales. ParameterSpec scales are 100 USD for amount and 12 hours for wait. Standardized Fisher traces rescale by these units; Tables in the main text report raw (unstandardized) diagonals unless noted.

Synthetic centre comparison. Table 3 lists offline synthetic metrics at the shared centre. Absolute Fisher scales are not comparable to live; the table is a methodological twin under the two-scale stencil.

Table 3: Offline synthetic centre $\theta_\star = (250, 8)$.

Node	H (nats)	$\mathrm{tr}(I)$	I_{amt}	I_{wait}
request_type	1.097	5.15×10^{-4}	8.20×10^{-7}	5.14×10^{-4}
needs_escalation	0.692	3.53×10^{-4}	2.55×10^{-6}	3.51×10^{-4}
impact	1.098	8.17×10^{-5}	1.31×10^{-6}	8.04×10^{-5}

B Artifact checklist

- `examples/jev_ticket_sensitivity/run.py` — offline + live entrypoints; stencil shrink.
- `examples/jev_ticket_sensitivity/paper_export.py` — CSV/JSON/PNG exporter.
- `papers/typesafe-ticket/data/*.csv` — committed summaries (no API keys, no full traces).
- `papers/typesafe-ticket/figures/*` — entropy, Fisher diagonals, synthetic grid, stability status.
- Capability refusals for system/action Fisher are intentional analysis-object boundaries.

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